IRAS Research Data Management Guidelines

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Preface

This book serves as a guideline for research data management at the Institute for Risk Assessment Sciences (IRAS) of the Faculty of Veterinary Science. It outlines and defines a set of principles regarding research data and provides practical handlers for researchers in accordance with the regulations established in the Utrecht University (UU) Policy Framework for Research Data Management and the Research Data Management Guidelines developed by the Faculty of Veterinary Medicine. These guidelines can be used as a reference together with existing UU policies, faculty guidelines and/or procedures already established by research groups.

Part I

Introduction

The current chapter describes:

- Chapter 1: how this book is structured.
- Chapter 2: why good research data management is important.
- Chapter 3: where you can find support.

1 Structure of this book

The IRAS Research Data Management guidelines are structured according to the Research Data Life Cycle (Figure 1.1). This cycle can be broadly divided into three phases: before (i.e. planning and designing), during and after research. Research data management (RDM) activities and actions take place during all three phases.

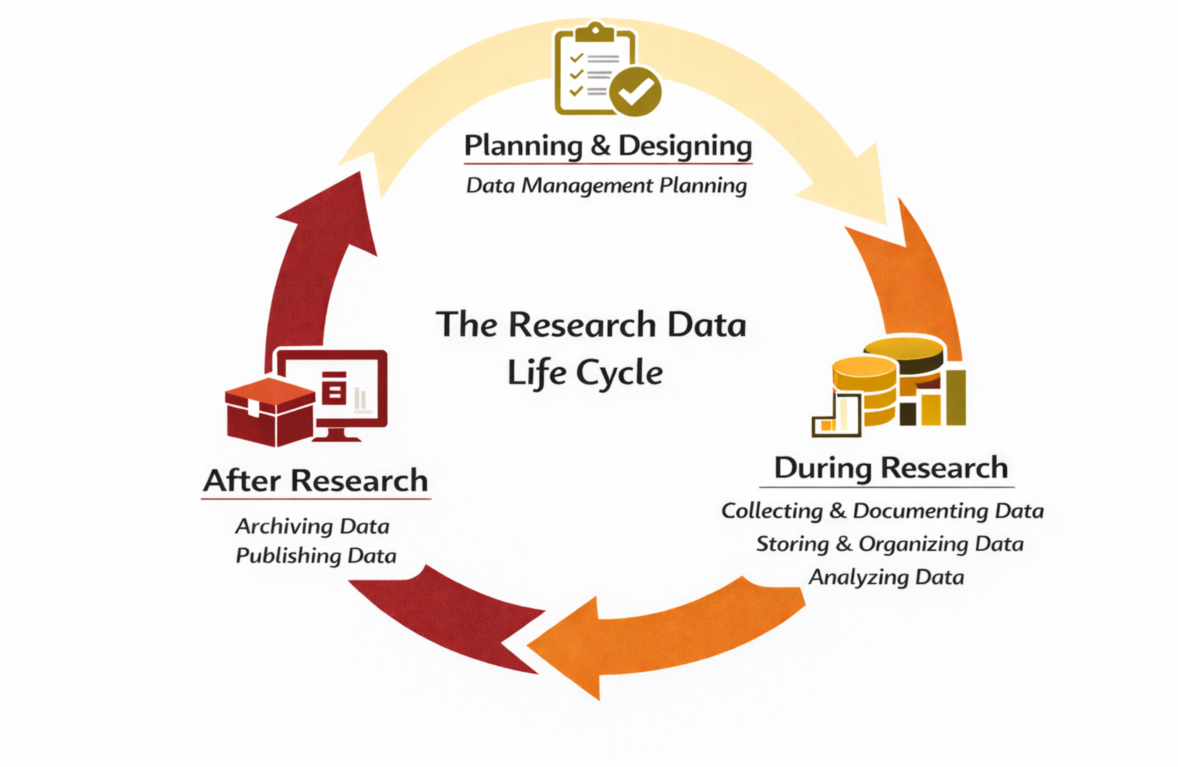

Figure 1.1: The Research Data Life Cycle

2 Importance of good research data management (RDM)

RDM refers to how you handle, organize, and structure your research data throughout the research process. Good RDM is essential for ensuring the integrity and validity of research, facilitating reproducibility and promoting collaboration. It also helps protect sensitive information and comply with regulations. Key reasons why good RDM is important include:

- **Ensuring data quality and security:** proper RDM prevents data loss, protects against unauthorized access, and ensures that research results are reliable and valid.
- **Increasing efficiency:** organizing, documenting, and structuring data saves time and resources by making it easier to find, understand and use data. Not only for others, but also for your future self.
- **Enabling reproducibility and validation:** well-managed, documented data allows other researchers to verify findings, contributing to the overall reliability and validity of scientific knowledge.
- **Allowing for reuse and collaboration:** making data understandable for others facilitates collaboration by enabling researchers to easily share and access data, accelerating scientific discovery.
- **Compliance and integrity:** good RDM ensures compliance with legal (e.g. GDPR when working with personal data), ethical, and funder requirements, and adheres to scientific integrity standards.
- **Improving visibility and impact:** making data FAIR (Findable, Accessible, Interoperable, Reusable) increases citation potential and the overall impact of research.

Proper RDM covers the entire data life cycle, from initial planning and collection to storage, analysis, archiving, and sharing. It is essential for effective Open Science and FAIR practices.

2.1 Open Science

In 2017, in order to accelerate and improve the realization of research results and their societal impact, the UU decided to make the transition to Open Science. Open Science is about increasing the reuse of research, and making sure that publicly funded research is accessible to all. Key to achieving this is adhering to the FAIR principles: ensuring the findings and data behind research results are findable, accessible, interoperable, and reusable (Bruce and Cordewener (2018)).

2.2 FAIR principles

In line with the UU's commitment to Open Science, research data at IRAS is expected to follow the FAIR principles:

- **F - Findable:** data and metadata should be findable, preferably digitally.
- **A - Accessible:** data and metadata should be stored and maintained in such a manner that they are accessible and with clear terms of use.
- **I - Interoperable:** data and metadata data should be stored and maintained in such a manner that the data can be easily opened, exchanged and combined with other datasets.
- **R - Reusable:** data are licensed and described in a manner that facilitates its reuse by other users.

Guidance on how to make your research data FAIR is highlighted in the recommendations provided below. To learn more see the RDM Support guide on how to make your data FAIR.

3 Support

3.1 IRAS Data Management team

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3.2 UU RDM Support team

In addition to the data management team at IRAS, the UU's Research Data Management (RDM) Support team is there to support, advise and assist all researchers at UU with data management, data analysis, research software and code. A multidisciplinary network of data experts, software specialists and research engineers offer their expertise through training, tools, infrastructure, consultancy, and custom-made solutions at every stage of a research project. The support they provide complies with the FAIR principles: findable, accessible, interoperable and reusable. That way, it contributes to Open Science.

> RDM Support workshops and guides
>
> To learn more about how to manage your research data and software, the UU's RDM Support team offers several trainings and workshops, as well as walk-in events. For a complete event agenda, together with information on trainings and workshop, please refer to the RDM Support website.
> Reading guides on diverse RDM topics can also be found on the RDM Support website. The same goes for tools that may be useful.

3.3 Other support services

Besides the IRAS Data Management team and RDM Support team, the following support services exist at UU:

- Research Support Office (RSO) of the Faculty of Veterinary Medicine: each faculty has an RSO. The RSO supports researchers throughout the year in the entire process of acquiring grants. For example: advice, information, reading, examples, budget assistance, data management, ethical questions, administrative data, etc.

- Privacy Officers of the Faculty of Veterinary Medicine: if you are starting a new project or research involving the use of personal data, exchanging data with a third party, or have questions about the secure storage of data or GDPR, please contact the Privacy Officers. You could also contact them for questions about the safe and responsible use of AI.

3.4 Contact

In case you have questions about research data management, please reach out!

> Contact details
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> IRAS Data Management team: iras_dm@uu.nl
> RDM Support team: info.rdm@uu.nl
> RSO of the Faculty of Veterinary Medicine: rso-dgk@uu.nl
> Privacy Officers of the Faculty of Veterinary Medicine: privacy-dgk@uu.nl

Part II

Planning and designing

The current chapter describes:

Chapter 4: responsibilities of everyone involved in the research project.

Chapter 5: why you should write a DMP and how to do that.

Chapter 6: why you should estimate the costs that will be associated with data management and how to do that.

4 Responsibilities

The **project leader** (e.g. principal investigator) is responsible for …

Depending on the project requirements, **project members** (e.g. researchers, PhD/MSc/BSc students, data managers, project managers, research assistants) should:

- sign a confidentiality agreement in case the project requires working with personal data;
- write a data management plan (DMP);
- …

5 Data management plan (DMP)

RDM Support has instructions for developing a DMP. At this page several questions on a DMP are answered, such as the ones below.

Questions and answers

- **What is a DMP?** A DMP is a formal document you develop at the start of your research project which outlines all aspects of managing your data, both during and after your project.
- **Why do you need a DMP?** Data management is all about making your work efficient and creating more value for your data for yourself and others, during and after your research, for instance by working towards FAIR (findable, accessible, interoperable, reusable) data.
- **How to write a data management paragraph for a grant proposal?** When writing a research grant proposal, funders, such as NWO, will often ask for a data management paragraph.
- **How to write a DMP?** When writing your own DMP, you can:
  - Use DMPonline to write your DMP: DMPonline contains templates from many different funders (e.g., NWO, ZonMw and ERC) as well as the institutional templates from participating universities. The template offered by Utrecht University is tailored to the requirements which are laid down in the University Policy Framework for Research Data And approved by both NWO and ZonMw.
  - Get advice or feedback: if you are in need of advice or want your DMP reviewed, you can contact Research Data Management Support anytime, or click "Request feedback" in DMPonline.
  - Need more information or an introduction? Follow the online training Learn to write your DMP or join the workshop Quick start to Research Data Management and ask all your questions.

The use of DMPonline is further explained at the following page: https://www.uu.nl/en/research/research-data-management/tools-services/tool-to-create-your-dmp-online.

5.1 Our recommendation

Our recommendation is to write a DMP at the start of your research project and use the applicable template in DMPonline. This contains several examples you can use for your project. If you have any questions or want to have your DMP reviewed, please contact RDM Support as outlined above. During onboarding this will be explained to new staff.

6 Costs

Research data management will likely incur financial costs. In particular, the costs of data storage, archiving and publishing can be high. In some cases, these costs may not be covered by the university, and thus they will need to be obtained through funding or via other routes during the planning phase of the project. Luckily, most funders consider the costs for data management eligible for funding.

RDM Support has developed a guide to estimate the costs of data management for each research phase and research activity. The costs of research data management will be a lot lower when you plan well ahead, instead of applying it retrospectively, both in terms of time and money.

Check out the RSO intranet page for help with the funding and grant acquisition process.

Part III

During research

The current chapter describes:

Chapter 7: what tools the UU offers to collect research data, how to use them and what to pay attention to in terms of data management.

Chapter 8: how best to store and organize your research data during your research project.

Chapter 9: why documenting your data is important and how to do this.

Chapter 10: where best to analyze your data and how to use GitHub for version control.

7 Collecting data

7.1 Formdesk

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7.2 Informed consent

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7.3 Qualtrics

If you want to create a survey that will collect personal data, the UU has one option and that is Qualtrics. Qualtrics is free of charge and can be used to set up surveys, collect data, distribute and analyze responses. Issues such as privacy and security are well taken care of. For further information please refer to the Qualtrics intranet page of the UU. It tells more about privacy, rules of conduct, non-standard UU modules in Qualtrics and more.

You can log into the Qualtrics website with your Solis-id (or UU email address), Solis-password and two-factor authentication. Once you have logged into Qualtrics, you can create a new survey via the Projects page. The Project page shows all surveys you have created (or that have been shared with you). You can collaborate on surveys with others by sharing them.

Appendix B offers recommendations and tips and tricks for building a Qualtrics survey in a FAIR and structured way, while keeping data quality and privacy in mind. This may take some investment upfront, but will pay off in the end.

8 Storing and organizing data

8.1 Storing data

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8.2 Organizing data

8.2.1 Folder structures

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8.2.2 File naming conventions

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8.2.3 Data formats

There is a lot of diversity in data formats. When possible, it is important to choose for future-proof data formats (i.e. open, standardized formats) to be able to continue reading, displaying and processing research data in the future.

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9 Documenting data

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10 Analyzing data

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Part IV

After research

The current chapter describes:

Chapter 11: how best to archive your research data after your research project.

Chapter 12: how best to publish your research data after your research project.

11 Archiving data

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12 Publishing data

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Part V

Data security and privacy

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13 Legal instruments and agreements, policies and laws

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14 Handling personal data

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15 Data security

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16 ...

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References

Bruce, R., and B. Cordewener. 2018. "Open Science Is All Very Well but How Do You Make It FAIR in Practice?" *LSE Impact of Social Sciences Blog.* London School of Economics & Political Science. https://blogs.lse.ac.uk/impactofsocialsciences/2018/07/26/open-science-is-all-very-well-but-how-do-you-make-it-fair-in-practice/.

Appendix A

List of mentioned tools

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Appendix B

Qualtrics tips & tricks

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